

POAMS: LECTURE 1

INTRODUCTION & HISTORICAL BACKGROUND I

1. General introduction

Welcome to the first in a series of lectures on Normal realism and POAMS, presented by Dr. Anthony Osborne. The lectures are designed to be accessible to the largest possible audience.

This lecture series seeks to encapsulate the ideas and output of a close collaboration between Viv Pope and myself, which lasted for over 25 years, from 1984 until Viv's death in 2012, as well as further developments in this work until the end of November 2021. POAMS is an acronym for **'The Pope-Osborne Angular Momentum Synthesis'**, which provides a new approach to Newtonian mechanics, Relativity, physical phenomena at the quantum level and Physics in general. It is based on an observer-centred philosophy known as **Normal Realism**.

The collaboration between Viv and myself was probably unique in academia since our work crosses the modern-day academic divide of arts and science. It combines expertise in Philosophy (Viv) and Mathematics (myself) in the spirit of what is traditionally known as Natural Philosophy. Our ultimate goal was to produce a *synthesis*, that is a *conceptually economical, logically and philosophically coherent* re-appraisal of the apparently disparate theories just mentioned. As such, rather than being just another Physics *theory* designed to supplant existing theories, it is based on an entirely different outlook, using a very different *philosophy of physics*.

What is meant by a conceptually economical approach to Physics involves an appeal to *Ockham's Razor*. This is the principle which in essence states that when presented with alternative explanations of physical phenomena, we should always opt for the simplest, the one with the fewest possible causes, assumptions or variables.

These lectures will cover some of the history relating to POAMS, the philosophical basis of POAMS, uniform motion and the constant c , traditionally interpreted as the 'speed of light', time-dilation in uniform motion, relativistic kinematics and dynamics, the ultimately discrete nature

of physical phenomena, the Pope-Osborne Angular Momentum Synthesis, angular momentum and spin, orbital time-dilation leading to Schwarzschild space-time, and some aspect of astrophysics and cosmology.

At one level, much of the material which I will present in these lectures can be thought of as representing a novel and simpler approach to Einstein's Special Relativity, Physics at the Quantum Level, Newton's theory of gravitation and Einstein's General Relativity. However, on another level our work is seen as controversial since it depends on a fundamental shift in philosophy which has far-reaching consequences in the way that Physics is conducted. It not only challenges the academic establishment but ultimately will also change the way that ordinary people think about the material world around them.

People may well ask why should they spent a large amount of time examining our work and fundamentally changing their ideas. One very strong reason lies in practical science. For example, POAMS predicts that *spin* has a direct effect on weight in a way that is not predicted by Newtonian theory. These predictions are now experimentally verified beyond a reasonable doubt. For the rest of this lecture and the next, I want to give a potted history of Normal Realism and POAMS in order to put this lecture series in context.

2. Historical background and context I

The origins of the Normal Realist philosophy can be traced back to 1954, when Viv Pope, then a young telephone engineer, was first introduced to Albert Einstein's (1878-1955) Special Relativity at an astronomical society meeting in Swansea, UK. Finding Special Relativity as abstruse as it was reputed to be, he naively wrote to Einstein to ask how it was logically possible that observers in different states of relative motion endured at different rates, leading to the so-called 'clock' or 'twins' paradox, in which a pair of twins moving relatively to one another could end-up both older and younger than the other.

Although Einstein did not often reply directly to correspondents, Viv received a hand-written reply, explaining that there was no such paradox.

The only ‘paradox’ lay in the conventional but unjustifiable preconception that ‘the passage of time is the same everywhere’. This got Viv wondering about the assumption for that conventional precept. What else can *time* be – that is, *real* or *physical* time – other than simply what is measured by ‘clocks’ of any sort, as opposed to the tacitly assumed, utterly mysterious ‘universal’ time, or cosmological ‘God’s-time’? In that case, since the only means of reading distant and fast-moving clocks is by means of light signals, and since the periods of those light signals (the ultimate ‘clocks’) are extended with motion away from us and shortened by motions towards us (the well-known ‘Doppler effect’), then it seemed to him pure common sense that the passage of times registered by those distant and fast-moving atomic clocks have to be systematically different from those we read on our own observatory clocks and spectrometers in the way that Einstein had demonstrated. Studies involving geometrical diagrams and logical arguments with others finally convinced him that Einstein was correct in that there was no paradox.

Puzzling over the practical and philosophical implications of this led Viv to continue his studies and, eventually, to give up his occupation as a telecommunications engineer for an academic vocation as a lecturer, researcher and writer in the Philosophy of Physics. This was with special emphasis on an objective and critical study of the origins of the customary conceptions of space and time.

The first realisation that Viv had that he was to develop a version of Special Relativity differing from Einstein’s was when he tried to represent space, time and motion geometrically. He discovered that with a geometrical scale constant, a distance-time conversion factor, c , of 186,000 miles to the second, he could derive Einstein’s time-dilation formula on the basis of Pythagoras’ Theorem alone, without having the need for the traditional interpretation of c as the ‘speed of light *in vacuo*’. This validity of this discovery was verified by mathematicians and physicists at his local university, Swansea, Wales.

Meanwhile, Viv critically studied other widely held tenets of Physics. One of these was the ‘Einstein-Podolski-Rosen (EPR) paradox’, the notorious cause of disagreement between Einstein, as the founder of Special Relativity, and Neils Bohr (1885-1962), a leading exponent of Quantum Theory. In Special Relativity, nothing can travel faster than the ‘speed of light’, c . But this was seriously at odds with *instantaneous action-at-a-*

distance required by Quantum Theory. Viv's unconventional (some would say heretical) solution to this paradox was simple. It was, on the grounds of conceptual economy, to jettison the conventional interpretation of c as being a *speed* and treat it in the way he had done in constructing his Pythagorean facsimile of Einstein's time-dilation formula; *i.e.*, simply as a *constant ratio* between units of the conventional measures of observational length and time. This paradigm shift led to the development of his Pythagorean 'cone model' of space-time geometry. With the help of a teaching colleague, John Hopton, Viv devised his three-dimensional graphic depiction of the dual aspect of time, combining quantum instantaneity and relativistic time-delay as two aspects of the same model.

The philosophical implications of both interpreting c simply as a conversion factor and the double aspect of time illustrated by the 'cone model', are radically different from any that might be drawn from Einsteinian Relativity, being more in line with the Relativistic Philosophy, or *Phenomenalism*, of Einstein's philosophical mentor, Ernst Mach (1838-1916). These philosophical implications were systematically explored by Viv in cooperation with a fellow student, G. Arwyn Evans, on a degree course in Philosophy at the University of Wales, Bangor. This was with the help and support of a former telecommunications colleague, Alan M. Smart. It involved a critical analysis of the use of language, along the lines pioneered by the eminent philosopher Ludwig Wittgenstein (1889-1951), of our classical conceptions of *distance*, *time* and *motion*. This led to a new paradigm of scientific philosophy, which they named **Normal Realism**. However, when Viv submitted these ideas as a thesis for a Master's Degree at Bangor, his thesis was rejected by the examiners because it crossed too many academic boundaries. 'Philosophers', he was told, 'should not try to tell physicists what they should and shouldn't do'. The eminent philosopher Sir Karl Popper (1902-1994) was later to have said that it was not Viv that had failed, but his examiners.

The discussions that led to the philosophical articulation of Normal Realism had also involved long dialogues in correspondence with the eminent philosophers, Sir Alfred Ayer (1910-1989), Gilbert Ryle (1900-1976) and Rom Harré (1927-2019), as well as Sir Karl Popper. There was also correspondence and, eventually, a meeting, between Viv and the

eminent scientist, Professor Sir Hermann Bondi (1919-2005), clinching their agreement over the redundancy of Einstein's interpretation of c as a 'speed *in vacuo*' and the validity of a much-simplified version of the foundations of Special Relativity based on the non-speed interpretation of that constant. Despite all this, Viv, with growing frustration, continually failed to find an academic outlet for his ideas for many years.

To talk now a little of my background, I graduated in Mathematics at The City University, London in 1976. During my final year there, I took a course in Special and General Relativity and became fascinated with both the underlying ideas as well as the mathematics. I continued at The City University in order to study for a doctorate, partly based on a new approach to General Relativity. This involved adjusting the equations of motion so as to erase the singularities predicted by General Relativity and thus avoiding the bizarre predictions of 'Black Holes' and the 'Big Bang', for which there was no direct scientific evidence. I was awarded my doctorate on 'Gravitation & Dynamical Systems' in 1980. The preceding October, I had accepted the post of lecturer in Mathematics at the then University of Keele (now Keele University). At Keele, I presented my own course on Special and General Relativity to finalists in Mathematics for many years.

It was at the end of 1982 that Viv left a copy of his self-produced booklet on his derivation of the time-dilation formula and the 'cone model' in the Department of Mathematics. This was entitled, 'The Phenomenon of Time-dilation (a New Approach to Relativity Theory)'. After reading and digesting the ideas in this booklet, I was intrigued by Viv's work and so we arranged a meeting at Keele. The outcome of this meeting was the start, in 1984, of a collaboration between Viv and I, a rare meeting of minds between a free-range philosopher and a mathematician, which lasted for over 25 years.

As a first task in our new collaboration, I offered support in writing a joint paper with Viv, giving a fuller mathematical account of his bare treatment of time-dilation and his 'cone model'. In doing so, it made Viv's ideas more presentable to the wider scientific community. Initially, in order to find a publication outlet, we decided to present the paper as a new approach to Special Relativity, which interprets c as simply as a conversion factor, leading to the time-dilation formula and the 'cone model'. In doing so, we

played-down any consideration of the underlying philosophy and the ramifications of such a paradigm shift. In this way our first joint paper was published in a mathematics education journal in 1987, having been accepted in 1985:

‘A new approach to Special Relativity’, N. V. Pope & A. D. Osborne, *The International Journal of Mathematical Education in Science and Technology*, Vol 18, No 2, (1987) pp. 191-198.

The publication was generally well-received, but only from the point of view of a teaching aid; the same reason that Hermann Bondi maintained for interpreting c as a conversion factor, as opposed to the ‘speed of light’. Not surprisingly, the Citations Index of that time described our paper as ‘a modelling approach to relativity’. The reviewer had failed to see that the ‘new approach’ had profound philosophical implications for the conduct of Physics.

Incidentally, a couple of years before this, the Head of the Department of Mathematics invited the Department of Philosophy at Keele to oversee Viv’s philosophical input to our collaboration, for a doctoral thesis. This was declined on the grounds that no-one in the Philosophy Department ‘felt competent to assess the mathematical-science aspect of the work’. Nevertheless, a rare interdisciplinary collaboration between Arts and Science had been established in the Mathematics Department at Keele, albeit as an academically unclassifiable, administratively uncomfortable project.

During the second half of the 1980s, Viv and I continued our discussions on Normal Realism and the interpretation of c as a conversion factor. These philosophical (as opposed to mathematical) discussions involved the behaviour of light, the ultimate discreteness of physical measures and instantaneous action-at-a-distance.

During the early years of the 1990s, Viv and I attended a biannual conference on ‘The Physical Interpretations of Relativity Theory’ (PIRT) in London. In the September 1990 meeting, we jointly presented a long paper (nearly 30 pages) on ‘Instantaneous Relativistic Action-at-a-

Distance’, with Viv giving the philosophical background to our ‘New Approach’, including the main ideas of Normal Realism, and I filling in the mathematical details of Viv’s derivation of the time-dilation formula and the ‘cone model’. The paper contained a comprehensive discussion on the implications for action-at-a-distance. It also included a section on our view of the essential *discreteness* of observational measures. We argued that in true relativism, sequences of events have to be ‘atomised’ to allow for true communication.

The conference paper also contained the first intimation of Viv’s idea that the motion of particles, or macro-physical bodies of any sort, has to be essentially *orbital* and that orbital behaviour is governed by the *holistic* conservation of angular momentum. Any interaction a body *A* may have with any other body *B* is immediate and reciprocal. If *A* is perturbed in some way, then the effect which is traditionally labelled as ‘gravitational’ on *B* and *B*’s inhibiting effect on *A*’s action, traditionally called ‘inertial’ are instantly balanced, in such a way that the overall angular momentum is conserved.

One of the participants of the September 1990 PIRT conference was Professor H. Pierre Noyes from the Stanford Linear Accelerator Center, U.S.A. He expressed a particular interest in our presentation and paper and suggested the journal *Physics Essays* as a possible outlet for our work. Subsequently, we submitted a paper to this journal and, with Noyes’ recommendation, it was accepted for publication. Our paper,

‘Instantaneous Relativistic Action-at-a-Distance’, N. V. Pope & A. D. Osborne, *Physics Essays*, Vol 5, No 6 (1992), pp. 409-421.

duly appeared.

Viv and I both attended a PIRT conference for the second time in September 1992. Viv gave a talk on our work divided between ‘Relativity and Realism’ and ‘The Holistic Simultaneity of Distant Interaction’. This built upon our talk from the 1990 meeting and addressed questions of a mainly philosophical nature. I gave a talk on our work entitled ‘The Action-at-a-Distance Spectrum’. This was concerned with the ultimate discreteness of physical measures and presented a novel logical deduction of the Balmer-Rydberg formula, which classically speaking, gives the

frequencies of vibration of the electromagnetic field manifest in the spectral lines from chemical elements. This deduction is based on a minimal number of suppositions consistent with our Normal Realist standpoint.

In the next lecture, I will continue with my potted historical account of POAMS and Normal Realism.